Card Casters

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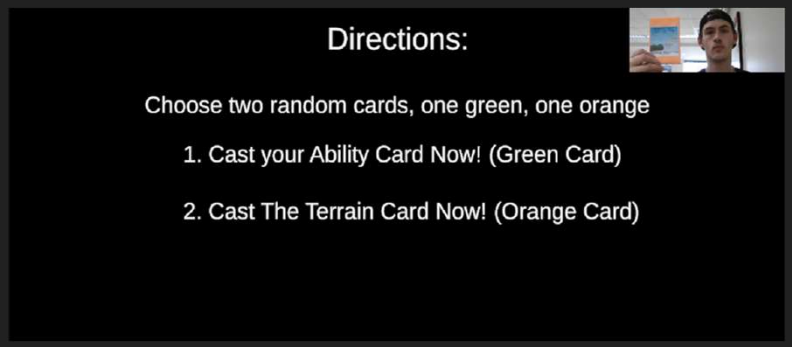
Artists: Kai Vore and Nate Oilar

Game Overview

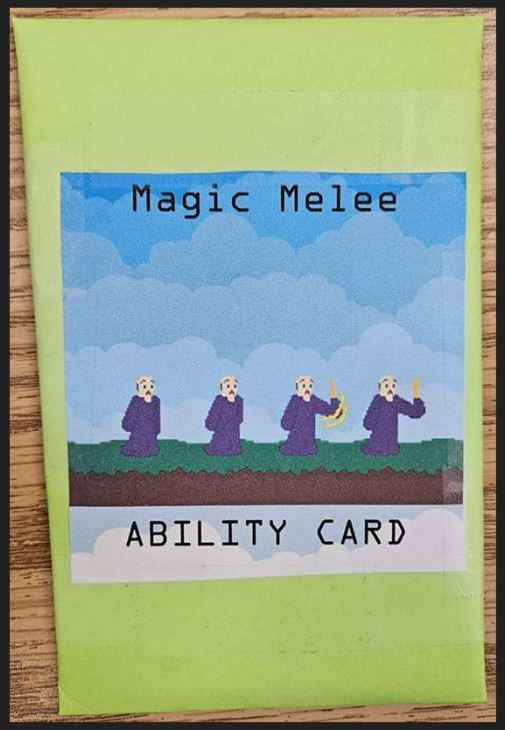
- Card Casters is a 2D Platformer game that utilizes AR in order to have an immersive experience and establish a difficulty for the game.
- The player plays as a wizard, who has tangible cards. At the start of each level the player is prompted to select one random terrain card (orange) and one random ability card (green).
- Each card has a certain size of terrain or ability that will aide the player in beating the level, while providing a sense of difficulty.

Vuforia and AR Explanation and Implementation

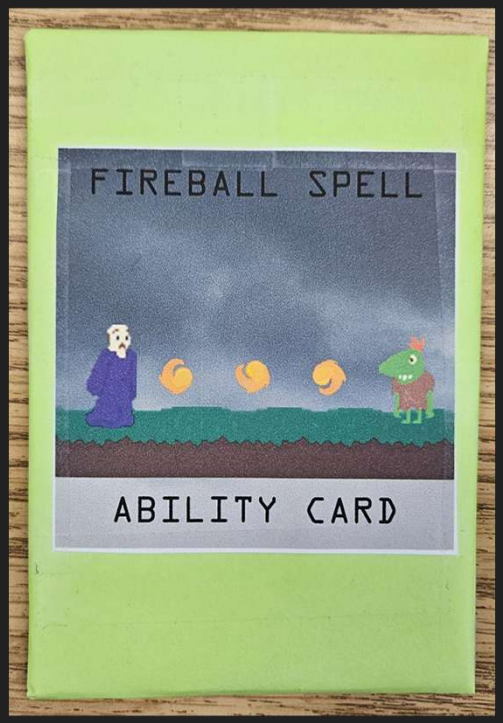


- Vuforia is a AR kit that was used to make this project.
- We used image targeting to scan the cards and upload the packages to unity.
- We set up a virtual camera that is in the corner of the screen that scans any cards that the player can use.

Card Overview

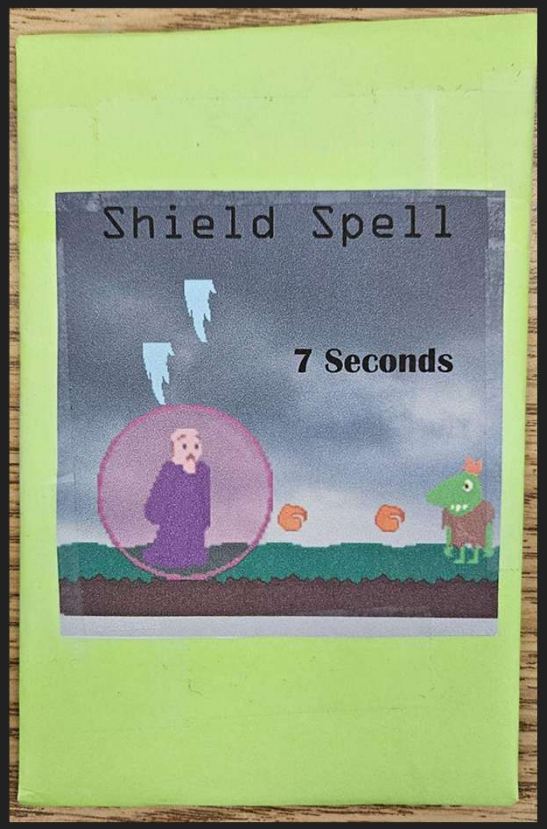


Once scanned at the beginning of the level, you have the ability to melee attack enemies, which is bound to the "E" key.



Once scanned at the beginning of the level, you have the ability to fire a projectile that kills enemies, which is bound to the "E" key.

Card Overview (Continued)

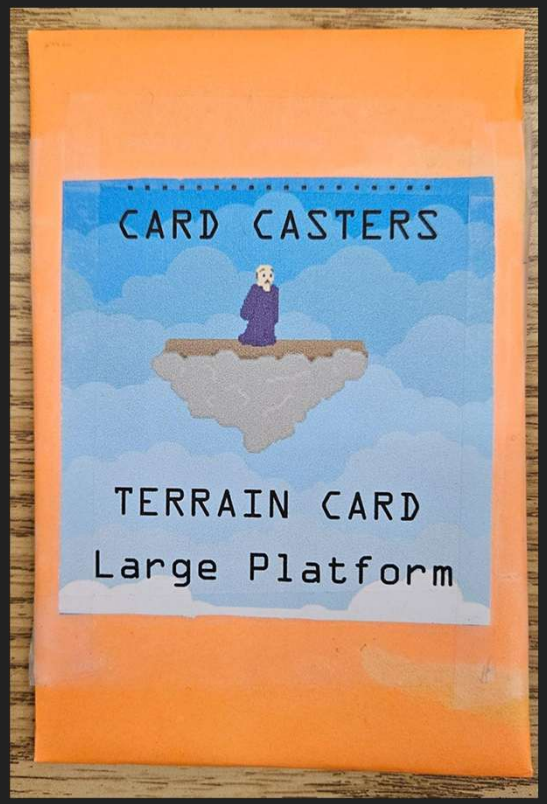


Once Scanned at the beginning of the game, you have the ability to activate a shield, you become invincible, and when you touch enemies, they die, this is bound to E.

This card is scanned at the beginning, but also scanned whenever it needs to be used. When scanned a small platform is spawned in front and above the player.

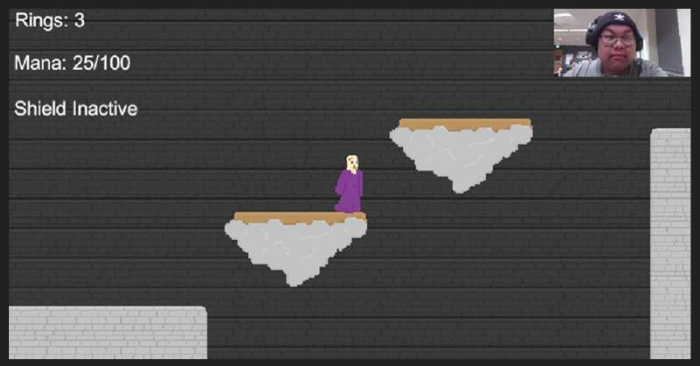
Card Overview (Continued)

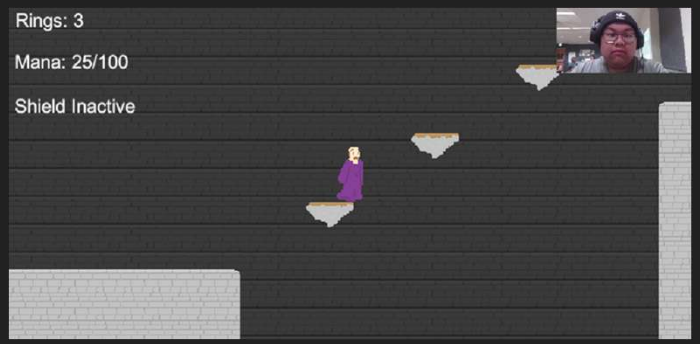
This card is scanned at the beginning, but also scanned whenever it needs to be used. When scanned a medium platform is spawned in front and above the player.



This card is scanned at the beginning, but also scanned whenever it needs to be used. When scanned a small platform is spawned in front and above the player.

How the Cards Impact Difficulty

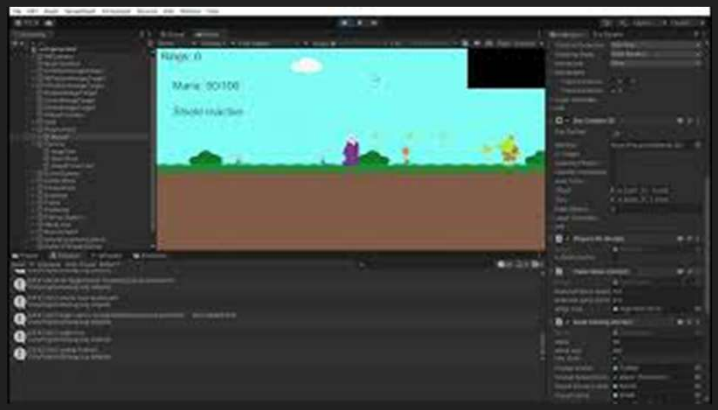


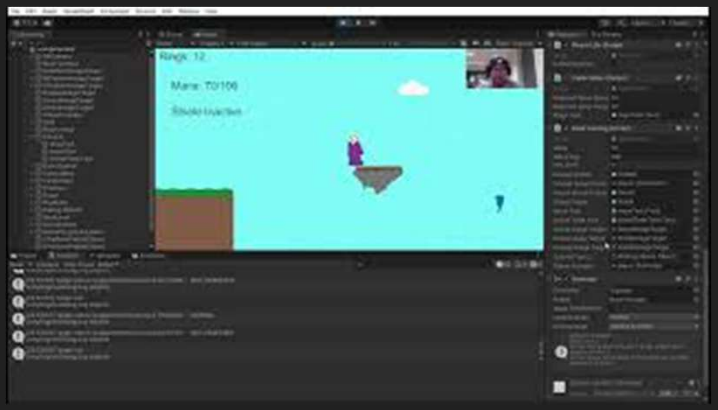


- The game is beatable no matter what cards you draw.
- Each Card adds a certain amount of aid towards the player.
- The size of the platforms makes it so that the player has more of a platform to stand on, which means less jumps.
- The different abilities also help the player survive and kill enemies.
- Each ability costs a certain amount of mana for balancing and so that the player can't spam abilities because mana is not rechargeable.

Physics Changes (Jaxon): Shooter and Falling Objects

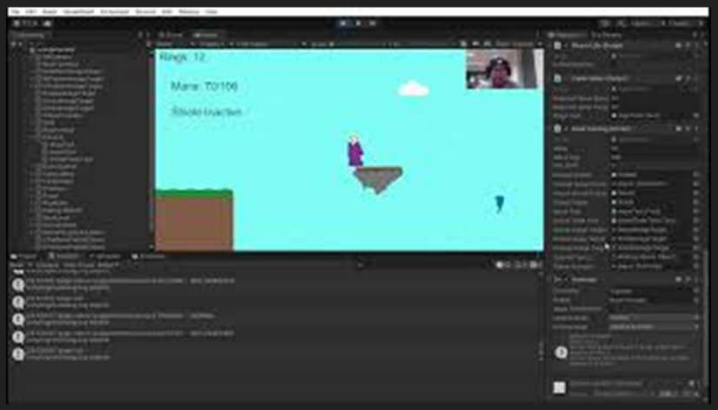
- Both the player and enemies are able to cast fireball ahead of them. The player's fireballs travel in a strict horizontal direction, while the goblin's fireballs can travel in any straight direction.
- Falling Objects are a kind of traps that are laid out throughout the level. This falling objects follow vertical waypoints and if they collide with a player, the player dies.





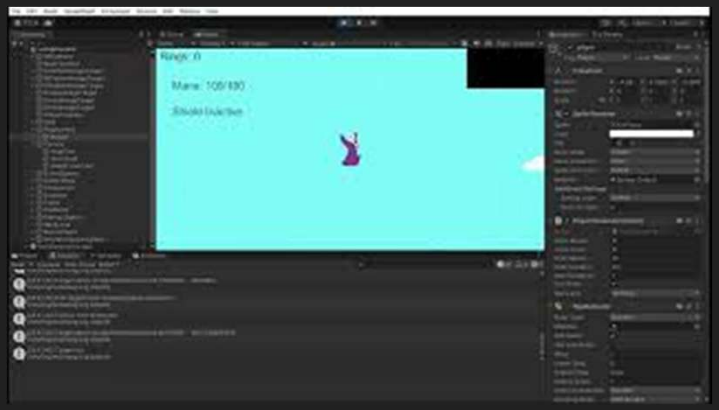
Physics Changes (Nate): Teleport and Waypoints

- Upon contact with a teleporter, the player will instantly be teleported to the corresponding teleporter. Returning through the other side is not allowed to prevent entering a loop upon teleport
- Waypoints are utilized for the moving platforms and falling obstacles to provide extra challenges



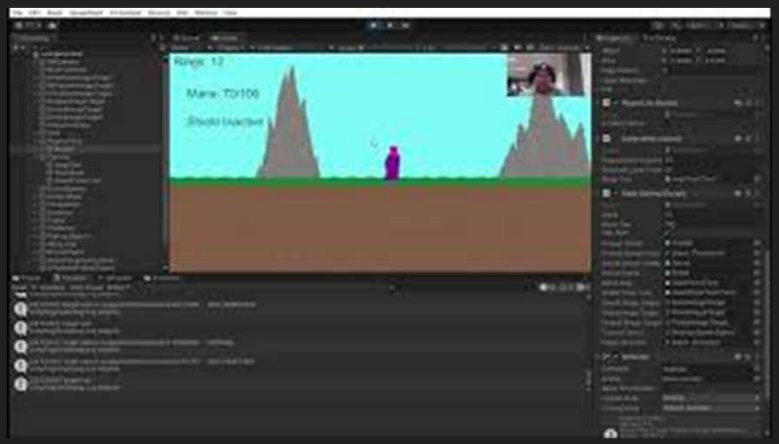
Physics Changes (Kai): Double Jump and Invincibility

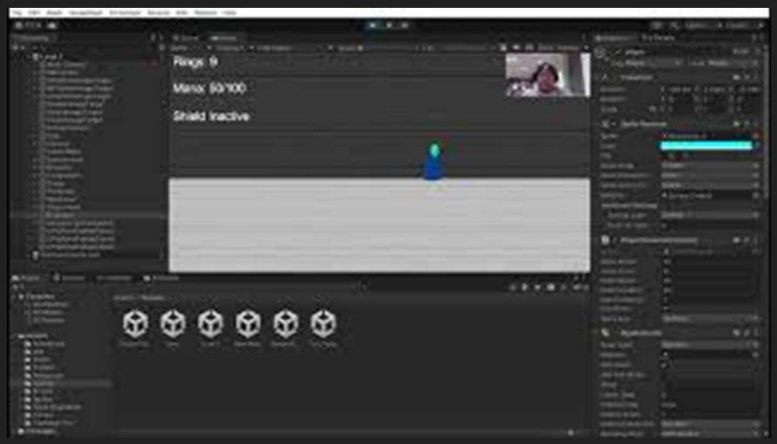
- The player is able to jump in midair, after either jumping once off of the ground, or running off of a platform
- Utilizing one of the spell cards, the player can gain temporary invincibility, where during the spell duration, the player can dispel any fireballs they make contact with, as well as defeating goblins upon contact



Physics Changes (Dylan): Jump and Speed Power Up

- Game objects that sit with in the level, when collided it changes the players movement



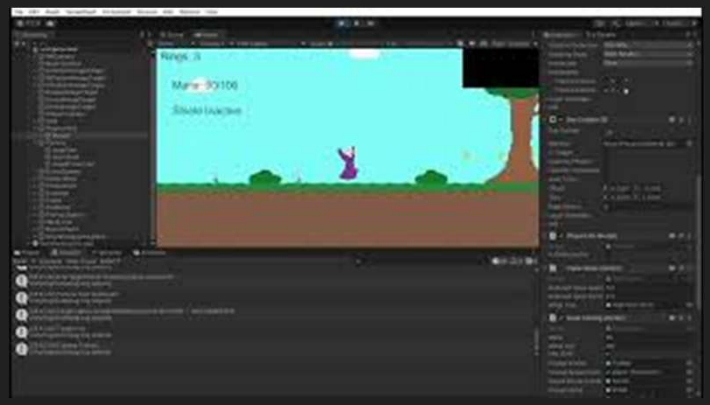


The jump power up will increase the player's jump force for a limited amount of time.

The speed power up will increase the player's speed for a limited time, and will revert to normal when the time runs out.

Physics Changes (Ty): Weight and Dash

The more coins you pick up the faster your base movement speed gets for the entirety of the life. The idea is the more rings you get, the lighter the character gets.



The dash ability is given at the start of the game. The player can press left shift and dash in a direction.

Our Experience and Takeaways

- This was our first game creation working in a team.
- Most of us were new to programming, digital art, Unity, Github, and definitely AR.
- Issues with Github ignoring Vuforia package.
- It become more efficient to work out the issues that arose, during the development, as a collective team.
- Collective creativity helped further the complexity of gameplay.